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Oefeningenreeks 4A nr. 14.10

$$\int \ln(x^x)dx = \int x \ln x dx$$
$$\begin{cases} u = \ln x \\ dv = xdx \end{cases} \Rightarrow \begin{cases} du = \dfrac{dx}{x} \\ v = \dfrac{x^2}{2} \end{cases}$$
$$= \frac{x^2 \ln x}{2} - \int \frac{x^2}{2x} = \frac{x^2 \ln x}{2} - \int \frac{x}{2} dx = \frac{x^2 \ln x}{2} - \frac{1}{2} \int xdx = \frac{x^2 \ln x}{2} - \frac{x^2}{4} + C$$

References